

Progress Report: LeGrad-Inspired Temporal Explanations for Speech Emotion Recognition

gradient_based_speech_xai

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1 Project Goal

The project studies speech explainability for emotion recognition on IEMOCAP. The planned pipeline compares:

1. SpeechXAI segment-level explanations.
2. A LeGrad-inspired temporal attribution method based on gradients with respect to Transformer attention maps.

Both methods will be converted to ranked time intervals in seconds and evaluated using the same deletion protocol. For each audio sample, selected regions are replaced by silence while preserving the original audio length. The main metric is the confidence drop for the original predicted class:

$$\text{drop} = p_{\text{original}}(\hat{y}) - p_{\text{masked}}(\hat{y}).$$

2 Repository Setup Completed

The repository has been scaffolded around the diagnostic-first workflow described in `skill.md`. The main additions are:

- `README.md`: project overview, setup instructions, notebook order, and expected outputs.
- `pyproject.toml`: editable-install package metadata and dependencies.
- `configs/default.yaml`: model name, paths, sample rate, SpeechXAI top-k values, random trials, and output file names.
- `data/README.md`: expected local IEMOCAP metadata format.
- `results/README.md`: placeholder for generated outputs.
- `src/speech_xai_project/`: reusable project helpers.
- `scripts/`: command-line runners for stable notebook logic.
- `notebooks/01` through `notebooks/09`: one notebook per project phase.

3 Reusable Code Added

The helper package `speech_xai_project` currently contains:

File	Purpose
<code>audio.py</code>	Audio loading, resampling, duration calculation, metadata discovery.
<code>config.py</code>	YAML config loading and project-relative path resolution.
<code>model.py</code>	SpeechBrain classifier loading and waveform classification helpers.
<code>intervals.py</code>	Score normalization, interval selection, random intervals, token-to-time conversion.
<code>masking.py</code>	Silence masking and total masked-duration calculation.
<code>legrad.py</code>	LeGrad-inspired attention-gradient relevance aggregation.
<code>evaluation.py</code>	Confidence-drop and deletion-result data structures.
<code>plotting.py</code>	Waveform plotting and interval highlighting helpers.

4 Notebook Plan Created

Nine notebooks were created to match the intended implementation order:

1. `01_inference_iemocap.ipynb`: model inference on IEMOCAP.
2. `02_attention_access_test.ipynb`: attention map access diagnostics.
3. `03_attention_gradient_test.ipynb`: attention-gradient diagnostics.
4. `04_legrad_temporal_explanation.ipynb`: LeGrad-inspired relevance intervals.
5. `05_speechxai_explanations.ipynb`: SpeechXAI segment explanations.
6. `06_unified_interval_masking.ipynb`: common interval format and silence masking.
7. `07_deletion_evaluation.ipynb`: confidence-drop evaluation.
8. `08_qualitative_examples.ipynb`: report-ready qualitative examples.
9. `09_final_results_summary.ipynb`: final tables and plots.

5 Step 1 Work Completed

Step 1 is the classifier inference phase. The following work has been done:

- `01_inference_iemocap.ipynb` was updated to load metadata from `data/metadata.csv` when available.
- If no metadata file exists, the notebook discovers audio files under `data/iemocap`.
- The notebook now raises a clear error when no audio is available instead of failing later at an unclear indexing operation.
- `scripts/run_inference.py` was added as a command-line equivalent of the Step 1 inference workflow.

- The runner writes `results/predictions.csv` once IEMOCAP audio and required dependencies are available.

The expected metadata format is:

```
audio_id,audio_path,true_label
Ses01F_impro01_F000,data/iemocap/path/to/file.wav,neutral
```

6 Corrected Quantitative Evaluation

The quantitative deletion protocol was updated after the initial scaffold. The old fixed-percentage masking budgets are no longer the project protocol. Because SpeechXAI naturally produces word-level explanations, SpeechXAI now defines the masking duration.

For each audio sample and each $k \in \{1, 2, 3, 5\}$, the top- k SpeechXAI words are selected and their total duration X is computed. SpeechXAI masks exactly those word intervals. LeGrad masks its highest-scoring time bins whose total duration equals X . The random baseline masks random time bins with the same total duration X , repeated over multiple random trials. This keeps the classifier, audio sample, masking operation, and amount of removed audio fixed across methods; only the selected regions differ.

The deletion-results table should therefore include:

```
audio_id, k, method, masked_duration, original_label,
original_confidence, masked_confidence, drop, relative_drop,
prediction_flipped, random_trial
```

7 Notebook Execution Results

The notebooks were executed using:

```
python -m nbconvert --to notebook --execute --inplace
--ExecutePreprocessor.timeout=300
--ExecutePreprocessor.kernel_name=python3 <notebook>
```

The command `jupyter nbconvert` was not available as a command shim, but `python -m nbconvert` worked.

Notebook	Result	Reason or Notes
01_inference_iemocap.ipynb	Failed	No IEMOCAP audio or <code>data/metadata.csv</code> .
02_attention_access_test.ipynb	Failed	<code>speechbrain</code> is not installed.
03_attention_gradient_test.ipynb	Passed	Placeholder diagnostic notebook executed.
04_legrad_temporal_explanation.ipynb	Passed	Placeholder explanation notebook executed.
05_speechxai_explanations.ipynb	Passed	Placeholder SpeechXAI notebook executed.
06_unified_interval_masking.ipynb	Passed	Placeholder masking notebook executed.
07_deletion_evaluation.ipynb	Passed	Placeholder evaluation notebook executed.
08_qualitative_examples.ipynb	Passed	Placeholder qualitative notebook executed.
09_final_results_summary.ipynb	Failed	<code>results/deletion_results.csv</code> does not exist yet.

8 Validation Already Run

The following validation checks have been run:

- Python compilation for `src` and `scripts`.
- JSON parsing for all nine notebooks.
- Notebook execution via `python -m nbconvert`.
- Step 1 smoke test with `python scripts\run_inference.py --limit 1`.

The Step 1 smoke test reached the intended data blocker and reported that no audio files were found under `data/iemocap` and no `data/metadata.csv` was present.

9 Current Blockers

Real experimental progress is currently blocked by missing local resources and dependencies:

- `torchaudio` is not installed.
- `speechbrain` is not installed.
- No IEMOCAP audio is present under `data/iemocap`.
- No `data/metadata.csv` exists.
- `results/deletion_results.csv` cannot exist until inference, explanations, masking, and deletion evaluation have run on real audio.

10 Next Steps

1. Install project dependencies, especially `torchaudio` and `speechbrain`.
2. Add IEMOCAP audio under `data/iemocap` or create `data/metadata.csv`.
3. Re-run `01_inference_iemocap.ipynb` and confirm that `results/predictions.csv` is generated.
4. Re-run `02_attention_access_test.ipynb` and identify where the `wav2vec2` encoder lives inside the `SpeechBrain` wrapper.
5. Only after attention maps and attention gradients are confirmed, implement the full LeGrad-inspired explanation notebook.

11 Development Rule Added

The project behavior in `skill.md` was updated so that each notebook created or modified for a project step must be executed before the step is considered complete. If execution is blocked by missing data, dependencies, or credentials, the blocker and failed command must be reported explicitly.